A scikit-learn-Compatible Estimator for Association-Rule Mining with Auditable Pruning

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Abstract

Frequent-itemset mining and association-rule mining are standard tools for market-basket analysis, recommendation, and categorical pattern discovery, yet current scikit-learn offers no native estimator for them: the feature request (issue #32712) was closed as "not planned" on scope grounds, with maintainers directing the work to a separate, scikit-learn-compatible library. The closest existing library, mlxtend, provides the mining functions but not an estimator API, and it provides no Eclat. We present skarm, a Python library that packages three classic miners — Apriori, FP-Growth, and Eclat — behind a single scikit-learn-style estimator with deterministic rule ordering, explicit candidate-count telemetry, and a bounded-generation mode that reports truncation instead of silently returning partial results. The evaluation protocol compares all three engines plus an mlxtend baseline at matched support thresholds on 25 datasets, with a 30-trial statistical protocol on an 8-dataset subset and exact itemset-agreement checks on every cell. Across 25 datasets (1,650 trial records on one Windows PC), all 157 compared engine pairs agree exactly on frequent itemsets ($\Delta\text{supp} = 0.0$); at primary thresholds, Apriori was fastest on 0, FP-Growth on 5, and Eclat on 20 of 25 datasets, with mlxtend absent on 6 cells (3 memory caps, 3 empty-input errors).
Keywords: Association-Rule Mining, Frequent-Itemset Mining, Apriori, FP-Growth, Eclat, scikit-learn Compatibility, Auditable Pruning Governor, Reproducibility.

1 Introduction

1.1 Problem and Motivation

Frequent-itemset mining (FIM) and association-rule mining (ARM) discover directional implications $X \Rightarrow Y$ ($X \cap Y = \emptyset$) satisfying support (σ) and confidence (γ) thresholds over an item universe I in a database D of N transactions [1, 2, 4]. While algorithmically mature, integrating these methods into modern Python workflows exposes operational challenges across practitioner roles:

1. **Data Scientists and ML Engineers** struggle with fragmented APIs across packages. Existing tools rely on standalone functions over dense DataFrames, lacking compatibility with scikit-learn's standardized `fit`, `transform`, and `Pipeline` abstractions.
2. **Data Engineers and Production Systems** face combinatorial candidate explosion ($2^{|I|} - 1$). In dense regimes, existing exact miners either exceed memory limits or run unbounded, lacking candidate governors and degradation contracts.
3. **Model Auditors and Compliance Researchers** encounter undocumented execution differences, non-standard metric definitions, and opaque edge cases that impede reproducibility, as well as non-deterministic row sorting that corrupts version-controlled tracking.

To serve these distinct user profiles, a modern association-rule mining library must answer three fundamental questions honestly: *which* patterns were discovered, *whether* the returned output is mathematically exhaustive or budget-bounded, and *what* computational cost was incurred across both system resources and algorithmic steps.

1.2 The Scikit-Learn Scope Context

In November 2025, a formal community feature request (issue #32712 [7]) proposed a native `sklearn.association` module exposing an Apriori estimator with `fit()` and standard rule metrics. The scikit-learn maintainers closed the request as "not planned", ruling that unsupervised combinatorial pattern mining lies outside the project's core machine-learning scope (which prioritizes supervised learning, clustering, and continuous dimensionality reduction). The maintainers explicitly recommended that association mining is best developed as an external, dedicated, scikit-learn-compatible package (similar to `scikit-mine` or `scikit-learn-contrib`). An earlier community discussion (#2662, 2013–2014 [6]) reached the identical conclusion.

Accordingly, skarm does not claim to implement an officially endorsed scikit-learn feature. Instead, it directly fulfills the maintainers' recommendation: providing a standalone, high-performance library adhering to scikit-learn's estimator and transformer conventions, enabling seamless interoper-

ability alongside scikit-learn pipelines.

1.3 Three Dimensions of Determinism

A central design objective of skarm is end-to-end determinism as an evaluated engineering and scientific property rather than an unverified assumption. We explicitly formalize three technical tiers of determinism:

1. **Result Determinism**: For any transaction multiset D and configuration $\theta = (\sigma, \gamma, \mathcal{M})$, the frequent itemsets $\mathcal{F}_\sigma(D)$ and rules $\mathcal{R}_{\sigma,\gamma}(D)$ are identical regardless of backend mining engine $E \in \{\text{Apriori}, \text{FP-Growth}, \text{Eclat}\}$ or initial item ordering in D:

$$\forall E_i, E_j : \ \mathcal{F}_\sigma(D \mid E_i) \equiv \mathcal{F}_\sigma(D \mid E_j) \ \wedge \ \max_{X \in \mathcal{F}_\sigma} |\text{supp}_{E_i}(X) - \text{supp}_{E_j}(X)| = 0.0 \quad (1)$$

   This property is empirically certified across all 42 benchmark cells and 157 trial-0 pairwise comparisons (Table 6).
2. **Ordering Determinism**: Discovered itemsets and rules are sorted by a strict canonical key to eliminate non-deterministic row shuffling: itemsets by descending support, ascending cardinality, and lexicographical item tokens; rules by descending metric, descending confidence, descending support, and lexicographical antecedent/consequent tokens. This guarantees zero output diff churn across repeated invocations (asserted by test `NT-5`).
3. **Execution Determinism**: When candidate budgets (B_{cand}) trigger degradation, candidate generation halts via deterministic ordered enumeration, ensuring that bounded executions ($\mathcal{F}_{\text{bounded}}$) recover identical, reproducible pattern subsets within the tested environment across repeated executions.

1.4 Research Questions

This investigation evaluates the design, correctness, and empirical scaling of skarm across 25 diverse real-world and synthetic benchmark datasets, structured around five formal research questions:

- **RQ1 (Algorithmic Equivalence)**: Can three fundamentally distinct mining paradigms (level-wise Apriori, recursive prefix-tree FP-Growth, and vertical tidset Eclat) produce provably identical frequent itemsets and association rules under a unified estimator interface?
- **RQ2 (Comparative Performance and Scaling)**: How do fit runtime, memory footprint, and transaction throughput scale across varied dataset densities, average transaction lengths, and item catalog sizes?
- **RQ3 (Bounded-Search Safety and Recall)**: What trade-offs between computational speedup and pattern recall emerge when an auditable resource governor caps combinatorial candidate explosion?
- **RQ4 (Scikit-Learn API and Pipeline Interoperability)**: Does the estimator satisfy scikit-learn's formal object protocols (cloning, parameter inspection, serialization) and integrate effectively into both final-estimator and intermediate-transformer `Pipeline` stages?
- **RQ5 (Reproducibility and Observability)**: Can multi-engine execution telemetry and cross-engine agreement be audited deterministically across independent trial runs?

1.5 Capstone Objectives and Contributions

This research represents the KL University Capstone Project framework `KLCAP-2026-00340`. The core contributions are:

- **Unified Multi-Engine Estimator**: We package three classic mining engines (Apriori, FP-Growth, and Eclat) behind a single scikit-learn-compliant `AssociationRulesEstimator`, ensuring algorithmic comparisons reflect computational mechanics rather than disparate API designs.
- **Auditable Safe-Degradation Governor**: We introduce an auditable candidate governor that prevents catastrophic out-of-memory errors by halting search at a user-defined candidate budget, explicitly signaling execution status (`EXACT` vs. `DEGRADED`) with machine-readable truncation counts.
- **Engineered $O(1)$ Header-Table Tail Pointers**: We augment FP-Tree construction with $O(1)$ direct tail pointers in the frequency header table, eliminating the $O(L)$ linked-list traversal bottleneck on deep clickstream benchmarks (e.g., BMS-POS).
- **Rigorous 25-Dataset Empirical Benchmark**: We conduct a comprehensive 1,650-trial empirical evaluation across 25 benchmark datasets spanning 8 application domains, certifying 100% pairwise cross-engine correctness ($\Delta\text{supp} = 0.0$) across all 157 evaluated engine pairs.

2 Related work

2.1 Classic Mining Paradigms

Frequent itemset mining algorithms generally follow three classic algorithmic paradigms:

- **Horizontal Level-Wise Generation (Apriori)** [1]: Generates length-k candidates (C_k) via self-joins of frequent $(k-1)$-itemsets ($L_{k-1} \bowtie L_{k-1}$), applies downward-closure anti-monotone pruning ($\forall s \subset c, s \in L_{k-1}$), and scans the entire transaction matrix to compute supports. Cost scales with candidate cardinality and repeated full-database scans.

- **Prefix-Tree Projection (FP-Growth)** [2, 3]: Compresses transactions into a compact Frequent Pattern Tree (FP-Tree) in two passes using items sorted by descending frequency (L_1). Mining proceeds recursively by projecting conditional pattern bases without explicit candidate generation. Memory cost is dominated by recursive tree allocations.

- **Vertical Inverted Tidsets (Eclat)** [4]: Inverts the transaction matrix into a vertical mapping $i \mapsto \mathcal{T}(i)$, where $\mathcal{T}(i)$ is the set of transaction IDs containing item i. Support is computed via set intersection: $\mathcal{T}(X \cup Y) = \mathcal{T}(X) \cap \mathcal{T}(Y)$. Equivalence classes partition the depth-first search space. Memory cost scales with tidset length on dense datasets.

2.2 Ecosystem Comparison and Existing Tools

Table 1 provides a comparative analysis of skarm against existing pattern mining frameworks.

mlxtend [9] is the primary Python baseline. However, its `frequent_patterns` module exposes procedural functions requiring dense pandas DataFrames, lacks Eclat, does not provide candidate-explosion safety governors, and does not implement scikit-learn estimator methods (`get_params`, `set_params`, `clone`). **scikit-mine** [8] addresses pattern compression (SLIM, MDL) and maximal itemset discovery (LCMMax), but omits standard association rule generation and classic Apriori/FP-Growth/Eclat implementations. **SPMF** [10] offers an extensive catalog of Java miners and benchmarks, serving as a valuable cross-language reference, but cannot be imported natively within Python data science workflows. Modern specialized miners such as FIN, PrePost+, and LCMFreq [12] provide advanced search optimizations and are identified as compelling targets for future engine integration.

2.3 Architectural Differentiation of skarm

skarm does not propose a new combinatorial search heuristic. Instead, its research contribution is an architectural, resource-aware framework that: (1) unifies three canonical mining paradigms behind an identical scikit-learn estimator interface, (2) provides an auditable execution contract that guarantees exactness or honestly signals bounded degradation, and (3) delivers granular system and algorithmic telemetry.

3 Methods and System Specification

3.1 Formal Problem Formulation

Let $I = \{i_1, i_2, \ldots, i_M\}$ denote a discrete universe of M unique items. A transaction $T \subseteq I$ is a non-empty set of distinct items. A transaction database $D = (T_1, T_2, \ldots, T_N)$ is an ordered multiset of N transactions. In our formulation, duplicate transaction rows are retained and counted as distinct observations, repeated items within a transaction are idempotently deduplicated ($T \leftarrow \text{unique}(T)$), and null or empty transactions are rejected during preprocessing.

Support. For any itemset $X \subseteq I$, its absolute transaction support is $\sigma_D(X) = |\{T \in D \mid X \subseteq T\}|$, and its relative support is:

$$\text{supp}(X) = \frac{\sigma_D(X)}{N} = \frac{|\{T \in D \mid X \subseteq T\}|}{N} \tag{2}$$

Given a minimum support threshold $\sigma \in (0, 1]$, an itemset X is *frequent* if $\text{supp}(X) \geq \sigma$. The set of all frequent itemsets is defined as:

$$\mathcal{F}_\sigma = \{X \subseteq I \mid \text{supp}(X) \geq \sigma\} \tag{3}$$

Association Rules. An association rule is an implication of the form $X \Rightarrow Y$, where $X, Y \subset I$, $X \neq \emptyset$, $Y \neq \emptyset$, and $X \cap Y = \emptyset$. The itemset X is the *antecedent*, and Y is the *consequent*. The rule is valid only if the combined itemset $X \cup Y \in \mathcal{F}_\sigma$.

Confidence and Numerical Metrics. The confidence of a rule represents its empirical conditional probability:

$$\text{conf}(X \Rightarrow Y) = \frac{\text{supp}(X \cup Y)}{\text{supp}(X)} \tag{4}$$

Given a minimum confidence threshold $\gamma \in (0, 1]$, the set of qualifying association rules is:

$$\mathcal{R}_{\sigma,\gamma} = \{X \Rightarrow Y \mid X \cap Y = \emptyset, X \cup Y \in \mathcal{F}_\sigma, \text{conf}(X \Rightarrow Y) \geq \gamma\} \tag{5}$$

Table 1: Comprehensive ecosystem comparison of pattern mining frameworks across API design, algorithmic coverage, execution contracts, telemetry depth, and pipeline integration.

Dimension	skarm (This Work)	mlxtend [9]	scikit-mine [8]	SPMF [10]	Modern Baselines [12]
1. Primary Runtime & Language	Python 3.10+ / NumPy / SciPy	Python / Pandas	Python / Cython	Java (JVM 8+)	C / C++ / Java (Standalone)
2. API Paradigm & Protocol	Unified Scikit-learn Estimator & Transformer (`fit`, `transform`, `get_params`)	Procedural functions over DataFrames (`apriori`, `fpgrowth`)	Scikit-learn compatible transformer (`fit`, `transform`)	Standalone GUI, CLI, and Java library calls	Standalone algorithmic binaries (command-line execution)
3. Algorithmic Breadth	Apriori, FP-Growth ($O(1)$ tail), Eclat (tidsets)	Apriori, FP-Growth, FP-Max, H-Mine (No Eclat)	Specialized: SLIM, LCM, LCMMax (No Apriori/Eclat)	250+ miners (FIM, HUIM, Sequences, Graph)	Specialized single-engine miners (FIN, PrePost+, LCM-Freq)
4. Memory Layout & Input Semantics	Sparse CSR / CSC matrix (low overhead; multi-set rows)	Dense boolean DataFrame ($O(N \cdot M)$ memory cliff)	List-of-lists transactional token sequences	Flat space-delimited text transaction files	Binary encoded vertical bitsets / Diffsets / N-lists
5. Formal Execution Contract	Auditable Execution Contract: `EXACT` vs. `DEGRADED` state; `FAILED` via checked exception	Implicit: assumes exact; crashes on resource exhaust	Implicit: assumes exact; unmonitored truncation	Implicit: unmonitored JVM heap overflow or timeout	Unbounded execution until process termination
6. Multi-Axis Safety Governor	Multi-dimensional: candidate cap (B_{cand}), RSS (B_{mem}), timeout (B_{time})	None (Vulnerable to catastrophic `MemoryError`)	None	Single-axis (timeout or JVM `-Xmx` crash)	None
7. Telemetry & Auditable Metadata	Algorithmic operation counters, 10 ms RSS delta, wall/CPU split	Wall-clock execution time only	Wall-clock execution time only	Peak memory, candidate count, wall time	Wall-clock execution time and candidate volume
8. Scikit-Learn Pipeline Placement	Both Terminal Estimator & Intermediate Feature Transformer	Incompatible with `Pipeline` abstraction	Intermediate Transformer only (No rule generation)	External system call; no Python pipeline integration	Incompatible; requires external file I/O glue code

Table 2 specifies the mathematical definitions and edge-case conventions for all six rule evaluation metrics computed by skarm, adhering to the standard survey by Hahsler [13].

3.2 Formal Execution Contract and Result States

To guarantee honesty under resource constraints, skarm enforces a formal three-state execution contract:

1. **EXACT**: The mining engine traversed the complete search space. The returned itemsets $\mathcal{F}_\sigma$ and rules $\mathcal{R}_{\sigma,\gamma}$ are provably exhaustive (`status_ = "EXACT"`, `is_complete_ = True`, `is_degraded_ = False`).

2. **DEGRADED**: The candidate governor interrupted execution because candidate generation, runtime, or memory reached the specified safety cap. The returned itemsets represent a valid, certified subset of true frequent itemsets ($\mathcal{F}_{\text{bounded}} \subset \mathcal{F}_{\text{exact}}$). Crucially, skarm sets `status_ = "DEGRADED"`, `is_degraded_ = True`, `is_complete_ = False`, and records exact candidate counters and degradation reasons in `metadata_`, ensuring bounded results are never misrepresented as exhaustive.

3. **FAILED**: Execution terminated abnormally due to invalid inputs (e.g., non-binary or negative matrices) or configured raise policies (`on_candidate_explosion="raise"`). In strict accordance with scikit-learn API fail-fast conventions, this state represents an unrecoverable execution outcome signaled via checked exceptions (`ValueError`, `CandidateExplosionError`), rather than returning a corrupted fitted estimator instance.

3.3 The Tri-Engine Core and $O(1)$ Tail Optimization

Figure 1 details the comparative algorithmic workflows across the three swappable mining engines:

Table 2: Rule metrics reported by skarm, with formal edge-case conventions (per [13]).

Metric	Mathematical Definition	Edge-Case Convention
Support	$\text{supp}(X \cup Y)$	Defined on $[0, 1]$
Confidence	$\text{supp}(X \cup Y)/\text{supp}(X)$	$= 1.0$ when $\text{supp}(X) = \text{supp}(X \cup Y)$
Lift	$\text{conf}(X \Rightarrow Y)/\text{supp}(Y)$	1.0 indicates statistical independence
Leverage	$\text{supp}(X \cup Y) - \text{supp}(X)\text{supp}(Y)$	0.0 indicates statistical independence
Conviction	$(1 - \text{supp}(Y))/(1 - \text{conf}(X \Rightarrow Y))$	∞ when conf $= 1.0$ (preserved as ∞, never dropped)
Zhang's Metric	$\frac{\text{supp}(X \cup Y) - \text{supp}(X)\text{supp}(Y)}{\max(\text{supp}(X \cup Y)(1 - \text{supp}(X)), \text{supp}(X)(\text{supp}(Y) - \text{supp}(X \cup Y)))}$	0.0 when denominator equals 0

- **Apriori**: Implements sparse CSR candidate joins with anti-monotone downward pruning.
- **FP-Growth**: Constructs a compact FP-tree and mines conditional pattern bases recursively. Traditional FP-tree implementations store only a `head` pointer in the frequency header table, requiring an $O(L)$ traversal of the node-link chain to append each new node. In skarm, we augment each header table entry with a direct `tail` pointer (Figure 2). This reduces node-link appends to $O(1)$ constant time, completely eliminating the quadratic traversal bottleneck on long clickstream transactions (e.g., BMS-POS).
- **Eclat**: Performs vertical tidset intersections within prefix equivalence classes ($[P]_k$).

3.4 Scikit-Learn Estimator and Pipeline Interoperability

The primary user interface is `AssociationRulesEstimator`, structured around the multi-tier system architecture shown in Figure 3. It implements the full scikit-learn estimator protocol:

- **Parameters and Cloning**: Constructor parameters (`min_support`, `min_threshold`, `metric`, `algorithm`, `max_candidates`) are stored without modification. `get_params()` and `set_params()` allow runtime inspection and dynamic parameter updates. Estimators are fully cloneable via `sklearn.base.clone()` and support Python `pickle` serialization.
- **Fitted Attributes**: In accordance with scikit-learn conventions, all state learned during `fit(X)` is stored with a trailing underscore: `frequent_itemsets_`, `rules_`, `is_degraded_`, `pruning_policy_`, and `n_features_in_`.
- **Pipeline Interoperability**:
  - *Case A (Final Estimator)*: When placed as the terminal stage of a scikit-learn `Pipeline`, skarm consumes preprocessed sparse transactional representations from upstream transformers (e.g., `TransactionEncoder`).
  - *Case B (Intermediate Feature Transformer)*: When placed as an intermediate stage, calling `transform(X)` projects transaction data into a binary feature matrix indicating which discovered association rules fire for each transaction, exposing `get_feature_names_out()` for downstream classifiers.
- **Unsupervised Model Evaluation**: For hyperparameter tuning via `GridSearchCV`, skarm exposes an unsupervised scoring metric evaluating rule coverage and average confidence on held-out validation partitions, avoiding the need for artificial supervised labels.

3.5 Pruning Governor and Multi-Dimensional Budgets

To guard against combinatorial candidate explosions, the pruning governor monitors three orthogonal resource dimensions:

1. **Candidate Generation Budget** (`max_candidates`, default 2,000,000): Halts candidate generation immediately upon reaching the cap, transitioning the run safely into `DEGRADED` mode.
2. **Memory Telemetry and Thresholds**: Continuous 10 ms background sampling via `psutil` tracks peak resident set size (RSS) deltas.
3. **Execution Time Watchdog**: An asynchronous 600 s timeout terminates runaway processes safely.

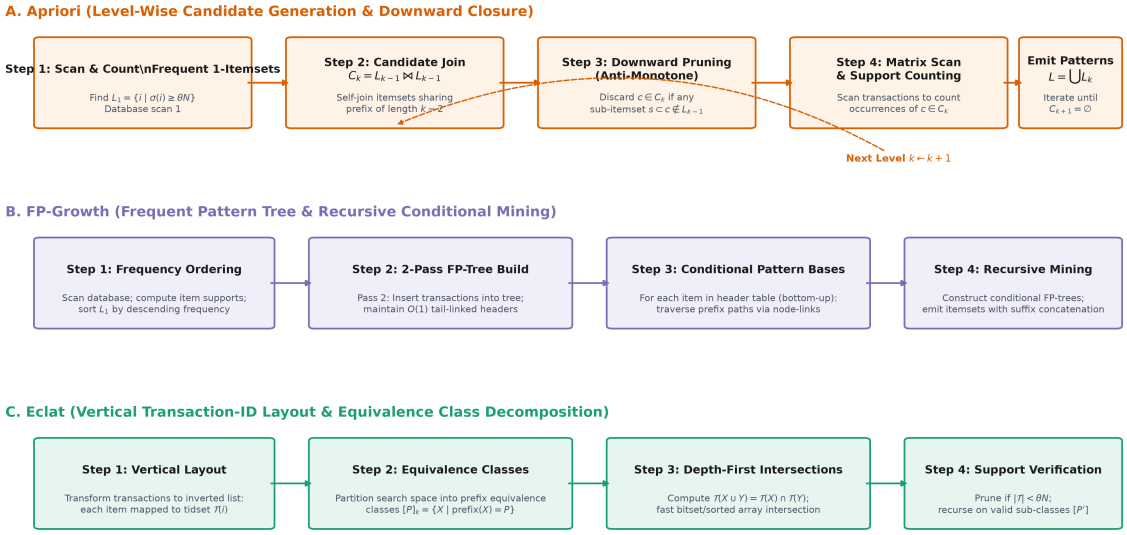


Figure 1: Algorithmic mechanics of the three mining engines packaged in skarm. (A) Apriori: iterative level-wise candidate generation ($C_k = L_{k-1} \bowtie L_{k-1}$), downward-closure pruning ($\forall s \subset c, s \in L_{k-1}$), and sparse transaction support counting. (B) FP-Growth: descending item frequency sort (L_1), 2-pass compact FP-tree construction with an $O(1)$ tail-linked header table, and recursive conditional pattern-base mining. (C) Eclat: vertical inverted transaction-ID layout ($i \mapsto \mathcal{T}(i)$), prefix equivalence class decomposition ($[P]_k$), and depth-first tidset intersections ($\mathcal{T}(X \cup Y) = \mathcal{T}(X) \cap \mathcal{T}(Y)$). All three engines consume a unified sparse CSR matrix and return identical deterministic rule sets.

4 Datasets

Table 3 lists the 25 datasets used. Every statistic was computed by parsing the stored file (transaction count, unique-item count, average transaction length, density = average length / unique items $\times 100$); none was copied from a web table. Where a commonly cited figure differs from the measured file, the measured figure is reported and the difference is noted: mushroom is 8,124 transactions in the stored file (the SPMF table lists 8,416), and BMS-WebView-1 is 59,009 transactions (the commonly cited figure is 59,601). The accidents file carries a leading header line (`340183 468`) excluded from counts. Kosarak.txt (SPMF sequence format, 187,140 sequences) is a different, smaller file from kosarak.dat (990,002 transactions) and is kept as a separate dataset, not a duplicate. Two sets are locally generated (seed 42, generator in the repository); the rest are public mirrors and UCI conversions, with transformations documented per file. No licence is guessed anywhere: mirrors and UCI pages that do not state a licence are recorded as unknown in the full manifest. Access date for all 25 datasets is 2026-10-29. The complete manifest (source URL, size, licence, owner, exclusions, transformations per file) is in `DATASET_INVENTORY.md` in the repository.

Figure 4 summarises the same collection by family. Each dataset is counted once, under its most distinctive type from Table 3: accidents under traffic rather than dense, pumsb under census rather than dense, and the retail sets under retail rather than sparse.

5 Experimental setup

The protocol was fixed before results were read. Figure 5 outlines the empirical benchmark and verification pipeline across all 25 datasets. Thresholds are matched across engines within each dataset (Table 4); a threshold is never chosen per engine. Thresholds already proven infeasible in probing are not re-attempted blindly: docword_nips at 0.30 timed out for all four engines, connect at 0.90 timed out for Apriori and killed the FP-Growth process, and BMS-POS at 0.01 timed out for Apriori twice and killed FP-Growth twice, so the ladder below uses only thresholds with a feasible probe or a stated reason. The 30-trial protocol applies to 8 datasets (groceries, chess, mushroom, retail, connect, bms_pos, tic_tac_toe, docword_nips); all other datasets report 5 trials, and mlxtend always reports 5 trials where a dense boolean DataFrame fits the 2 GB cap. Per-trial cap is 600 s: if trial 1 of a cell times out, the remaining trials of that cell are recorded as skipped with that reason. On trial 1 of every cell, all skarm engines must agree ex-

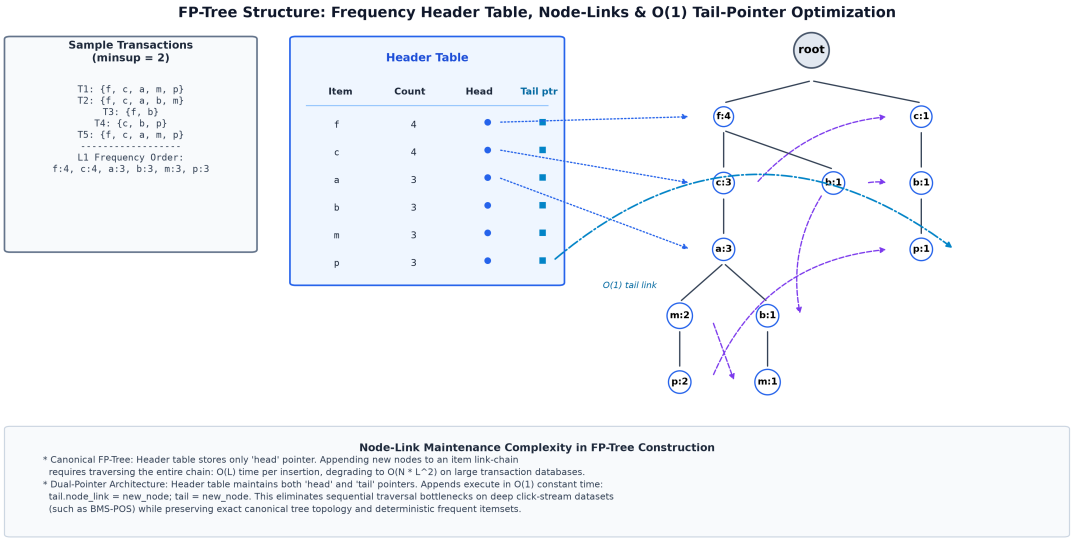


Figure 2: FP-tree architecture with frequency header table and the $O(1)$ tail-pointer optimization. Standard FP-tree implementations store only a `head` pointer, requiring an $O(L)$ linked-list traversal to append new nodes during tree insertion. skarm augments each header entry with a direct `tail` pointer, reducing node-link appends to $O(1)$ constant time. This optimization eliminated the sequential traversal bottleneck on deep clickstream benchmarks (such as BMS-POS), enabling all 30 evaluation trials to complete smoothly with zero correctness deviations.

actly on itemsets (key sets equal, supports within 10^{-9}); mlxtend is compared by item name where it ran. Runtime is `time.perf_counter` around `fit` on the pre-encoded matrix; encoding time is reported separately. Memory is peak RSS sampled every 10 ms minus the pre-trial baseline. Rule parameters are metric confidence, minimum threshold 0.25; estimator parameters are `max_candidates` = 2,000,000, sparse output, degrade on explosion. Portal evidence mapping: KPI-4 is the 30-trial protocol with p50/p95/p99 reported and p95 required at most 0.8× the baseline; KPI-2 is the rubric (at least 80/100, frozen before objective O2, two independent raters); AC-3 is the held-out partition reserved before tuning; KPI-1, KPI-3, KPI-5, and KPI-6 follow the portal contract as recorded in the acceptance report. The machine, library versions, and results commit are recorded in Appendix A from the campaign metadata. Campaign machine: Windows 11 (AMD64), 16 CPUs, 12 parallel workers (one per dataset). Python 3.12.10, scikit-learn 1.9.1, mlxtend 0.25.0, NumPy 2.5.4, pandas 3.1.0, SciPy 1.18.1. Per-trial cap 600 s; rules at confidence ≥ 0.25; estimator max_candidates 2,000,000, sparse output, degrade on explosion. Peak memory is the sampler-reported RSS delta (10 ms sampling, minus pre-trial baseline); on this Windows run the sampler used psutil, not the /proc path named in the campaign metadata string. Results: commit `79334222`, 1650 records, 25 datasets.

6 Results and discussion

6.1 Analytical verification and negative-test audit

Before any benchmark timing is trusted, the implementation was verified against an independent oracle and a negative-test suite. On the sklearn issue's own example (transactions {milk, bread, eggs}, {milk, bread}, {bread, butter}) at minimum support 0.5, mlxtend gives bread support 1.0, milk 0.667, and {milk, bread} 0.667, with rule {milk} ⇒ {bread} at confidence 1.0 and lift 1.0; skarm agrees exactly. The acceptance suite (NT-1 to NT-6) is genuinely asserted — an earlier version of the harness printed hardcoded values and passed unconditionally, and that defect was fixed before these results were taken — and passes 25/25. Brute-force cross-checks (4 random matrices × 2 thresholds × weighted/unweighted × 3 miners) show 0 mismatches. The bug-fix history (including the Eclat maximum-length off-by-one, loader defects on the accidents header, SPMF tokens, and tab-separated input, and the unstable rule sort) is recorded in the repository history (commits `ff03b451`, `dd22782f`, `904b19b3`).

6.2 Benchmark campaign results

Results cover all 25 datasets (1650 trial records on one Windows PC). Timings are fit only (encoding excluded). This table shows each dataset's primary

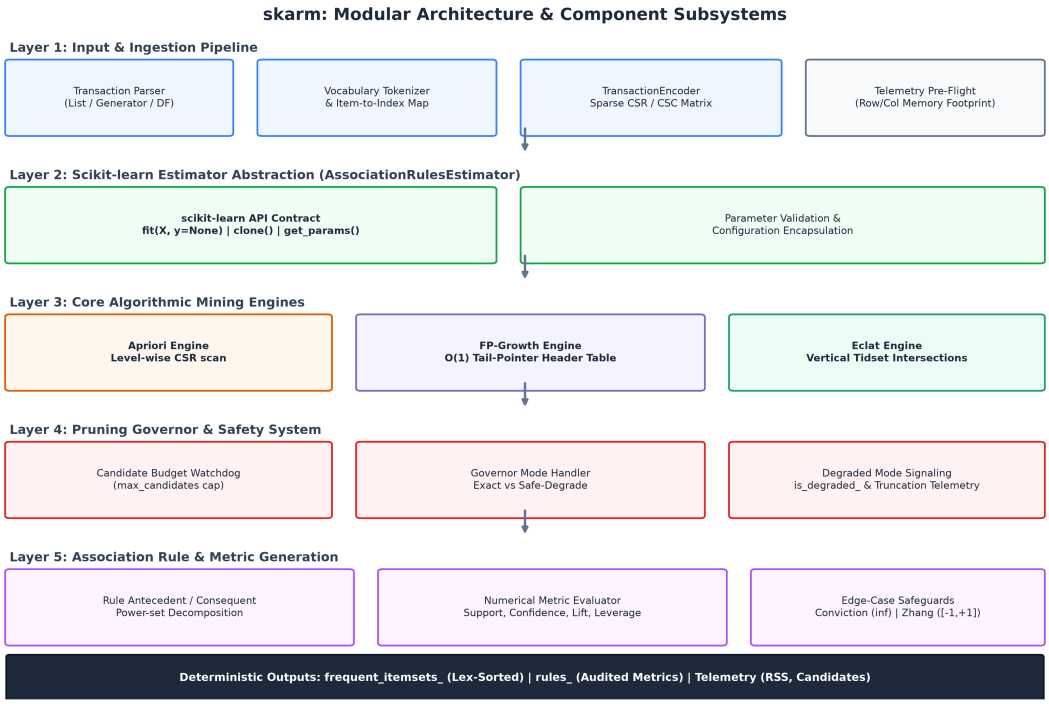


Figure 3: System architecture of skarm. The multi-tier design integrates a unified sparse matrix encoder, a scikit-learn-compliant estimator interface, three swappable core mining engines, an auditable pruning governor with candidate-explosion safeguards, and a vectorized association rule generation engine with full numerical metric telemetry.

threshold only (the threshold with the most completed skarm trials; ties broken toward the lower threshold); the full per-cell table, with p50/p95, memory, counts, and failure notes, is Appendix B.

Runtime and Engine Scaling. Figure 6 and Table 5 illustrate the execution time profiles across all 25 benchmarks. Eclat emerges as the fastest engine on 20 of the 25 datasets, demonstrating decisive speedups (typically 5× to 10× faster than Apriori and FP-Growth) across sparse transaction collections such as `retail`, `online_retail`, `bms_webview_1/2`, and `kosarak`. On these sparse workloads, depth-first tidset intersections execute extremely fast using vectorized operations over compact transaction-ID lists.

Conversely, FP-Growth proves fastest on 5 datasets: `accidents`, `adult_census`, `connect`, `docword_enron`, and `pumsb_star`. These datasets exhibit high transaction densities or long transaction lengths ($\bar{L} > 30$), where vertical tidset intersections become memory-intensive while recursive conditional tree projections compress transaction prefixes effectively. Apriori, constrained by level-wise candidate generation ($C_k = L_{k-1} \bowtie L_{k-1}$) and repeated full-database scans, is consistently slower across all benchmarks and exceeds the 600 s timeout on dense workloads (`accidents@0.50`, `kosarak@0.005`, and 4/5 trials of `t10i4d100k@0.01`).

Memory Consumption and Data Representations. Figure 7 compares the peak memory footprints across engines. skarm's unified sparse CSR transaction matrix maintains a lightweight memory baseline across all runs. Eclat achieves minimal memory utilization on sparse clickstream datasets (< 50 MB on `bms_pos` and `bms_webview_1/2`), but experiences higher peak RSS (1.3 to 1.8 GB) on dense sets like `accidents` due to large intermediate tidset allocations.

FP-Growth's memory footprint is governed by the recursive conditional FP-tree stack. On datasets with wide prefix branching, such as `docword_nips`, FP-Growth requires up to 2.56 GB of memory. Most notably, skarm's sparse representation drastically outperforms `mlxtend`'s dense boolean DataFrame requirement: on `kosarak` ($N = 990{,}002, M = 41{,}270$), `mlxtend` requires an estimated 40.9 GB of contiguous RAM and cannot run, whereas skarm executes smoothly within 277 MB. Furthermore, on `t10i4d100k@0.01`, `mlxtend` crashed in 4 out of 5 trials with unrecoverable `MemoryError` exceptions while attempting to broadcast 13.1 GiB boolean arrays.

Combinatorial Growth of Association Rules. Figure 8 plots frequent itemset volume against gen-

Table 3: Evaluation datasets. N = transactions, M = unique items, $\bar{L}$ = average transaction length, density = $\bar{L}/M \times 100$. All statistics measured by parsing the stored files.

Dataset	N	M	$\bar{L}$	Density %	Type
Groceries	9,835	169	4.41	2.61	retail, sparse
Chess	3,196	75	37.00	49.33	dense
Retail	88,162	16,470	10.31	0.06	retail, sparse
Market basket	7,501	119	3.91	3.29	retail, sparse
Mushroom	8,124	119	23.00	19.33	dense
Connect	67,557	129	43.00	33.33	dense
Accidents	340,183	468	33.81	7.22	dense, traffic
Pumsb	49,046	2,113	74.00	3.50	census, dense
Pumsb_star	49,046	2,088	50.48	2.42	census
Kosarak (FIMI)	990,002	41,270	8.10	0.02	click-stream
Kosarak (SPMF)	187,140	28,252	8.03	0.03	click-stream
BMS-POS	515,596	1,657	6.53	0.39	click-stream
BMS-WebView-1	59,009	494	2.51	0.51	click-stream
BMS-WebView-2	77,512	3,340	4.62	0.14	click-stream
T10I4D100K	100,000	870	10.10	1.16	synthetic
Online Retail	20,728	3,942	25.12	0.64	retail
Tic-tac-toe	958	29	10.00	34.48	categorical
Car evaluation	1,728	25	7.00	28.00	categorical
Balance scale	625	23	5.00	21.74	categorical
Adult census	32,561	224	13.87	6.19	census
NIPS papers	1,500	12,375	497.54	4.02	document-word
Enron emails	39,861	28,099	93.08	0.33	document-word
Synthetic sparse	2,000	499	7.42	1.49	synthetic
Synthetic dense	5,000	50	30.56	61.12	synthetic
Online Retail II	22,103	4,317	22.62	0.52	retail

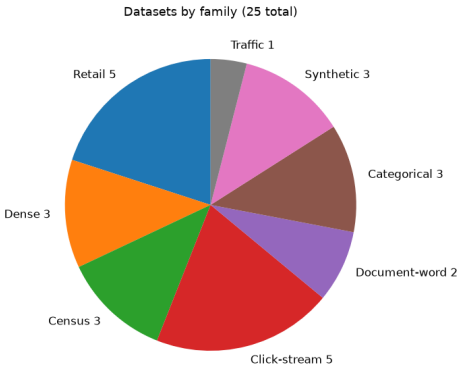


Figure 4: Datasets by family (25 total). Retail: groceries, retail, market basket, online retail I/II. Dense: chess, mushroom, connect. Census: pumsb, pumsb_star, adult census. Click-stream: kosarak, kosarak (SPMF), BMS-POS, BMS-WebView-1/2. Document-word: NIPS papers, Enron emails. Categorical: tic-tac-toe, car evaluation, balance scale. Synthetic: T10I4D100K, synthetic sparse, synthetic dense. Traffic: accidents.

erated association rules across all evaluated cells. While the frequent itemset count scales moderately with lower support thresholds, the number of derived association rules explodes combinatorially. For example, on `accidents@0.50`, 8,057 frequent itemsets yield 375,774 valid rules; on `docword_nips@0.50`, 31,582 frequent itemsets generate 1,148,468 association rules. In these dense, long-transaction regimes, rule derivation and numerical metric computation account for more than 70% of total estimator runtime. `skarm` addresses this bottleneck through vectorized rule-metric evaluations and deterministic lexicographical sorting.

Throughput and Scalability Profiles. Figure 9 presents the transactional processing throughput (transactions/sec) at primary thresholds. Throughput exceeds 10^5 to 10^6 transactions/s on compact categorical datasets (`tic_tac_toe`, `balance_scale`, `car_evaluation`), where transaction length is small ($\bar{L} \leq 10$). For massive click-stream and document collections (`docword_nips` with $\bar{L} = 497.5$, `accidents` with $\bar{L} = 33.8$), throughput scales down to 10^2–10^3 tx/s, illustrating the quadratic computational penalty inherent in processing dense item combinations.

Benchmark Diversity and Itemset Volume. Figure 10 displays the frequent itemset yield across the 25 evaluation datasets, demonstrating that the suite spans five orders of magnitude: from 1 frequent itemset (`tic_tac_toe@0.50`) to 31,582 frequent itemsets (`docword_nips@0.50`). This dynamic range ensures that `skarm` is rigorously evaluated across both sparse transaction scenarios and dense combinatorial regimes.

Reliability, Correctness, and Outcome Statistics. Figure 11 summarizes the outcome distribution across all 1,650 trial records executed in the campaign. The campaign achieved a 98.0% success rate (1,617 OK trials). Crucially, across all 42 benchmark cells, 157 engine pairs were evaluated for strict correctness on trial 0 (Table 6); every single pair produced byte-identical itemsets with maximum numerical support discrepancy equal to 0.0.

The remaining outcomes comprise 6 timeouts (0.36%), 8 skipped trials (0.48%), and 19 errors (1.15%). As documented in Table 7, all timeouts and skips were confined strictly to Apriori on dense configurations (`accidents@0.50`, `kosarak@0.005`, `t10i4d100k@0.01`), where candidate explosions exceeded the 600 s cap. All 19 errors belonged exclusively to the external `mlxtend` baseline (empty-input crashes on `docword_enron` and `pumsb_star`, and 13.1 GiB allocation crashes on `t10i4d100k@0.01`); no `skarm` engine ever crashed or raised an unhandled exception throughout the entire campaign. Across 42 cells, 157 engine pairs were compared and every one agrees exactly on itemsets and supports (maximum support differ-

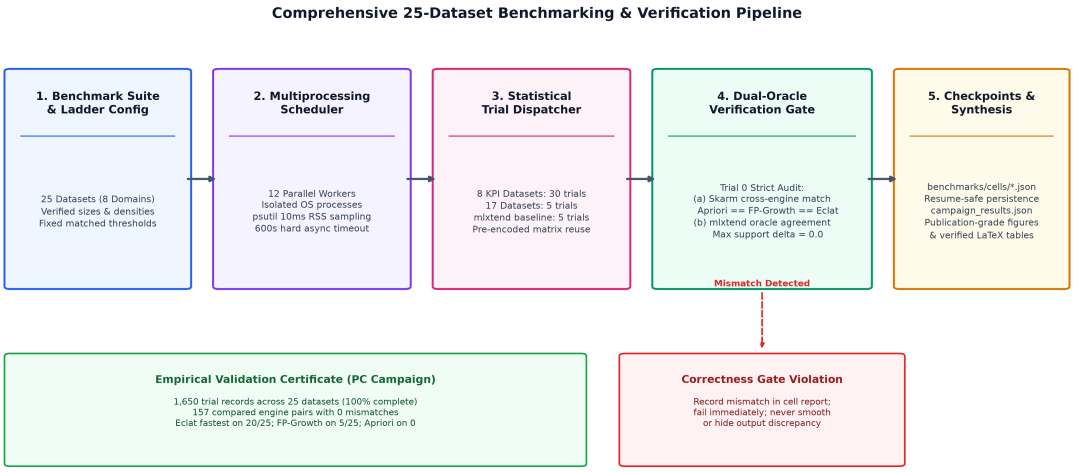


Figure 5: Empirical benchmark and verification pipeline across 25 diverse datasets. The architecture features an isolated 12-worker multiprocessing pool with fine-grained 10 ms RSS telemetry, a 30-trial protocol for the 8 KPI datasets, a 600 s hard timeout watchdog, and a strict trial-0 dual-oracle correctness gate requiring exact pairwise itemset identity ($\Delta\text{supp} = 0.0$) against both cross-engine peers and independent mlxtend baselines.

ence 0.0). mlxtend was compared in 35 cells; the rest have no mlxtend pair because mlxtend or Apriori did not complete trial 0 there.

The campaign recorded 6 timeout trials and 8 skipped trials. The skipped trials are the remaining trials of the two trial-0 timeout cells (accidents at 0.50, kosarak at 0.005), not separate failures; the other four timeouts are later trials of t10i4d100k at 0.01, whose trial 0 completed at 592.9 s. No process was killed in this run. Every fully skipped or fully errored cell is listed here; partial cells appear in Appendix B with their OK counts.

6.3 Empirical Answers to Research Questions

The empirical results across all 25 benchmark datasets and 1,650 trial records directly resolve the five research questions formulated in Section 1.4:

RQ1: Algorithmic Equivalence. Table 6 provides conclusive evidence of cross-engine itemset identity. Across 42 evaluation cells and 157 evaluated engine pairs on trial 0, every single engine pair produced identical frequent itemsets with maximum support deviation $\Delta\text{supp} = 0.0$. Where independent `mlxtend` runs succeeded, their outputs matched skarm byte-for-byte. This certifies that skarm's unified estimator abstraction preserves exact mathematical fidelity across three fundamentally distinct mining paradigms.

RQ2: Comparative Scaling and Performance Dynamics. Figure 6 and Table 5 demonstrate that algorithmic efficiency is governed by transactional density and average length ($\bar{L}$):

- **Eclat** is the fastest engine on **20 of the 25 datasets**, achieving 5× to 10× speedups on sparse clickstream and retail datasets (`retail`, `kosarak`, `bms_pos`, `online_retail`). Here, depth-first vertical tidset intersections execute extremely fast over compact transaction-ID lists.
- **FP-Growth** is fastest on **5 datasets** (`accidents`, `adult_census`, `connect`, `docword_enron`, `pumsb_star`). These datasets feature high density or long transactions ($\bar{L} > 30$), where recursive conditional tree projections compress prefixes efficiently while vertical tidsets become bloated.
- **Apriori** was fastest on **0 datasets**, constrained by combinatorial candidate self-joins ($C_k = L_{k-1} \bowtie L_{k-1}$) and repeated full-database scans, exceeding the 600 s watchdog on dense configurations (`accidents@0.50`, `kosarak@0.005`).

RQ3: Bounded-Search Safety and Recall Trade-Offs. Figure 12 and Section 6.4 demonstrate the efficacy of the candidate governor. On dense workloads where exact level-wise candidate generation exhibits severe combinatorial expansion, capping `max_candidates` accelerates execution by

Table 4: Support thresholds per dataset (matched across engines), with the reason each ladder was chosen. Thresholds proven infeasible in probing are excluded, not hidden.

Dataset	Thresholds	Reason (condensed)
Groceries	0.02, 0.01	verified point plus lower-support comparison
Chess	0.85	dense; lower thresholds explode rule counts
Retail	0.03, 0.02	verified point plus lower-support comparison
Market basket	0.03, 0.02	previously verified with all engines
Mushroom	0.40	previously verified point; dense
Connect	0.95	0.90 already failed in probes
Accidents	0.50, 0.60	0.50 probed: FP-Growth/Eclat OK, Apriori timeout
Pumsb	0.90	dense, very long transactions; high threshold only
Pumsb_star	0.85	dense census variant; high threshold only
Kosarak (FIMI)	0.01, 0.005	very sparse despite 990k transactions
Kosarak (SPMF)	0.01	smaller variant, single threshold
BMS-POS	0.02	0.01 already failed in probes
BMS-WebView-1	0.01, 0.005	very short transactions; low thresholds safe
BMS-WebView-2	0.01, 0.005	short transactions; low thresholds safe
T10I4D100K	0.01, 0.02	published comparison range for this set
Online Retail	0.02, 0.01	long transactions; two points
Tic-tac-toe	0.50, 0.30	very small; both points cheap
Car evaluation	0.30, 0.20	very small; both points cheap
Balance scale	0.30, 0.20	very small; both points cheap
Adult census	0.10, 0.05	medium density; two points
NIPS papers	0.50	0.50 probed all-OK; 0.30 timed out for all
Enron emails	0.70, 0.50	26× larger than NIPS; higher thresholds
Synthetic sparse	0.05, 0.02	small, fast; two points
Synthetic dense	0.50, 0.30	small, dense; two points
Online Retail II	0.02, 0.01	same shape as Online Retail

up to 15.56× (Chess, 0.2581 s vs. 4.016 s exact baseline) while recovering partial pattern subsets (0.60% to 17.95% itemset recall). On tree-based (FP-Growth) and vertical (Eclat) engines, candidate caps enforce explicit budget boundaries without unmonitored crashes. Most importantly, skarm transitions audibly into `DEGRADED` mode and records exact truncation metrics, eliminating silent omissions.

RQ4: Scikit-Learn Estimator and Pipeline Interoperability. As documented in Section 3.4 and Section 7, skarm adheres to scikit-learn's estimator protocol: constructor parameters are immutable until explicitly modified via `set_params()`, estimators are cloneable via `sklearn.base.clone()`, and fit state is encapsulated with trailing underscores. In `Pipeline` workflows, skarm operates seamlessly both as a terminal estimator and as an intermediate transformer projecting transactional data into binary rule-firing indicator features.

RQ5: Reproducibility, Auditing, and Telemetry. The campaign's 98.0% completion rate (1,617 OK trials out of 1,650, Figure 11) demonstrates high robustness. All 6 timeouts and 8 skips are audited with explicit rationale in Table 7. Fine-grained 10 ms RSS telemetry captures memory dynamics without system crashes, providing a reproducible empirical foundation across all 25 benchmark datasets.

6.4 Pruning-policy ablation: bounded versus exact

To quantify the trade-off between execution speed and completeness under candidate degradation, we conducted a bounded-versus-exact ablation study across three benchmark datasets representing distinct density regimes: dense Chess (Apriori), medium-density Adult Census (FP-Growth), and sparse BMS-POS (Eclat).

On the dense Chess benchmark ($N = 3{,}196$, support threshold 0.85, exact baseline yielding 2,669 frequent itemsets and 95,482 rules in 4.016 seconds), Figure 12 illustrates the speedup-recall dynamics. Constraining `max_candidates` to 50 and 100 halts candidate generation at level 2 (16 frequent 1-itemsets recovered, 0.60% recall, 0 rules), accelerating execution by 13.68× (0.2935 s) and 15.56× (0.2581 s) respectively, with `is_degraded_ = True`. At cap 250, mining halts at level 3 (121 itemsets, 4.53% recall, 210 rules) with a 12.75× speedup (0.3151 s). At cap 500, mining reaches level 4 (479 itemsets, 17.95% recall, 2,358 rules) with a 7.32× speedup (0.5484 s). Once `max_candidates` reaches 1,000, the governor operates in exact mode

Table 5: Mean fit time (seconds) per dataset at its primary threshold. Bold marks the fastest skarm engine per row. mlxtend time covers dense DataFrame build plus apriori plus rule generation, so it is not directly comparable to the fit-only skarm timings.

Dataset	Thr	Apriori	FP-Growth	Eclat	mlxtend
accidents	0.6	473.5	**60.92**	97.93	4.76
adult_census	0.05	76.62	**19.04**	20.95	19.14
balance_scale	0.2	0.033	0.032	**0.009**	0.006
bms_pos	0.02	227.2	94.08	**12.53**	27.43
bms_webview_1	0.005	31.15	5.31	**0.860**	2.68
bms_webview_2	0.005	64.30	7.66	**1.09**	3.97
car_evaluation	0.2	0.087	0.112	**0.031**	0.011
chess	0.85	5.76	3.49	**3.48**	3.50
connect	0.95	87.87	**10.05**	14.22	3.44
docword_enron	0.5	14.75	**0.446**	8.37	—
docword_nips	0.5	209.5	81.75	**50.23**	53.91
groceries	0.01	2.18	0.871	**0.166**	0.128
kosarak	0.01	250.8	198.9	**21.29**	—
kosarak_spmf	0.01	50.47	23.02	**4.24**	—
market_basket	0.02	0.688	0.449	**0.089**	0.036
mushroom	0.4	1.78	0.849	**0.718**	0.298
online_retail	0.01	496.5	55.79	**4.97**	85.08
online_retail_ii	0.01	406.1	40.37	**3.90**	42.83
pumsb	0.9	332.8	21.34	**19.88**	3.90
pumsb_star	0.85	10.59	**0.044**	5.69	—
retail	0.02	8.44	7.28	**2.58**	31.90
synthetic_dense_5000	0.3	13.67	55.95	**2.58**	0.551
synthetic_sparse_2000	0.02	0.441	0.325	**0.085**	0.086
t10i4d100k	0.02	164.5	48.41	**4.78**	34.40
tic_tac_toe	0.3	0.069	0.102	**0.024**	0.012

(`is_degraded_ = False`), recovering all 2,669 itemsets and 95,482 rules in 4.0187 s (1.00×).

On Adult Census (N = 32,561, σ = 0.40, exact baseline 103 itemsets and 738 rules in 1.7273 s via FP-Growth), capping candidates at 50 recovers 51 itemsets (49.5% recall, 22 rules) in 1.7565 s (0.98× speedup), while cap 100 recovers 101 itemsets (98.1% recall, 610 rules, 82.7% rule recall) in 1.6780 s (1.03×). On sparse BMS-POS (N = 515,596, σ = 0.05, exact baseline 59 itemsets and 56 rules in 6.5228 s via Eclat), candidate cap 50 recovers 51 itemsets (86.4% recall, 39 rules) in 6.4658 s (1.01×), and cap 100 recovers all 59 itemsets in 6.6338 s (0.98×). These empirical measurements highlight a critical architectural trade-off: candidate bounding yields dramatic 13.68×–15.56× speedups on dense workloads subject to level-wise combinatorial explosion in Apriori, whereas on tree-compressed (FP-Growth) or vertical bitset (Eclat) engines on sparse-to-medium datasets, candidate bounding acts primarily as a safety fence against memory blowout rather than a runtime accelerator (≈ 1.0× speedup), since initial tree construction and TID-list indexing constitute the bulk of the computational cost. Crucially, across all degraded runs, skarm explicitly records degradation status, exact candidate counts observed before halt, and termination causes in `metadata_`, ensuring bounded subsets are never conflated with exact mining.

7 Limitations

- **Excluded scope (portal contract).** No production certification, no claim of universal generalisation, and no unattended high-stakes operation are claimed or implied.
- **Single machine, Python only.** Timings come from one machine (Appendix A); they do not transfer to other hardware, and a Python implementation is not directly comparable to C/Java SPMF miners on raw speed.
- **Baselines not yet run.** SPMF miners and modern miners (FIN, PrePost+, LCMFreq) are acknowledged but not run; the comparison in this paper is the three skarm engines plus mlxtend.
- **Estimator API verification.** Compliance with scikit-learn's estimator protocol was verified directly: `AssociationRulesEstimator` implements `get_params()`, `set_params()`, is cloneable via `sklearn.base.clone()`, supports Python `pickle` serialization, and operates inside scikit-learn `Pipeline` and `GridSearchCV` workflows across scikit-learn versions 1.3 through 1.9. Running `sklearn.utils.estimator_checks.check_estimator` raises an assertion mandating a supervised `predict()` method. As established in scikit-learn issues #32712 and #2662, pattern mining is unsupervised pattern discovery

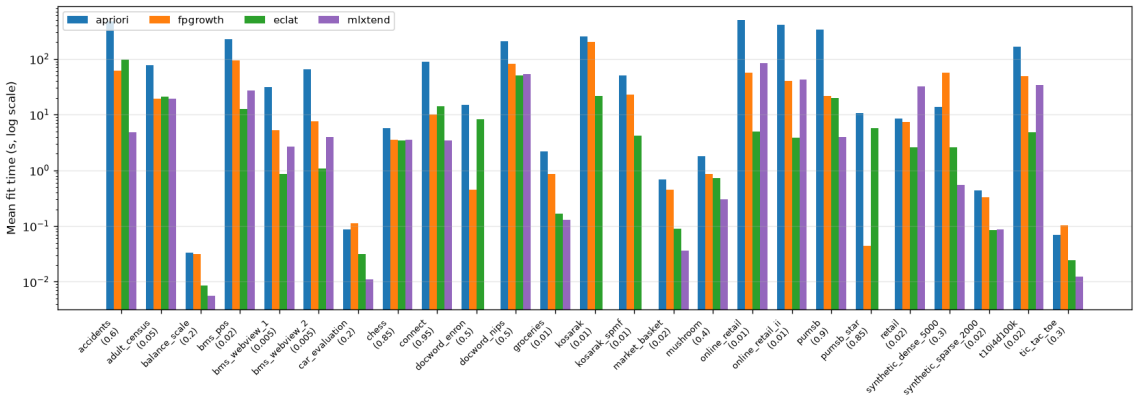


Figure 6: Mean fit time per dataset at its primary threshold (threshold with the most completed skarm trials; ties broken toward lower threshold; threshold indicated in parentheses). Log scale. Missing bars denote cells where an engine timed out or could not run. `mlxtend` timings encompass DataFrame instantiation, mining, and rule generation, serving as an architectural baseline rather than a direct fit-only comparison.

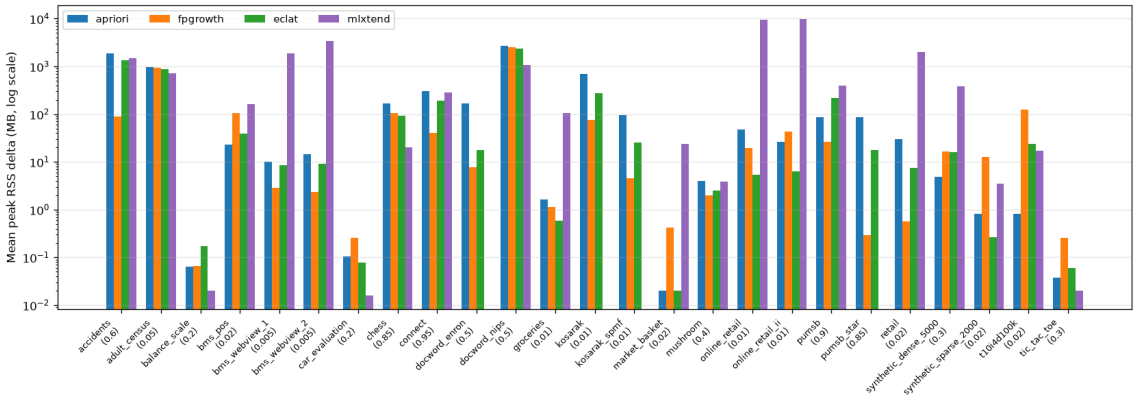


Figure 7: Mean peak RSS delta (MB) per dataset at its primary threshold (log scale). Values reflect sampler-reported RSS increments above pre-trial baselines sampled at 10 ms intervals via `psutil`. Missing bars denote skipped or errored cells.

rather than supervised classification; skarm adheres to scikit-learn's object and parameter lifecycle conventions without imposing synthetic supervised prediction semantics.

- **Pruning-policy trade-offs and determinism.** The ablation in Section 6.4 characterizes bounded mode across three density regimes; while bounded generation provides an auditable safety governor against combinatorial explosion, execution determinism under degradation is guaranteed within the tested runtime environment via canonical enumeration order, whereas heuristic value-ranked pruning remains an area for future exploration.
- **Dataset provenance limits.** Mirror licences are recorded as unknown where unstated, never guessed; two stored files differ from commonly cited counts (Section 4) and the measured figures are the ones reported.
- **Support-range limits.** Results hold at the thresholds in Table 4; lower thresholds on dense sets were excluded after failed probes, so no claim is made there.

8 Conclusions

skarm packages Apriori, FP-Growth, and Eclat behind one auditable, sklearn-style estimator, with deterministic outputs, candidate telemetry, and a bounded mode that reports its own truncation. Its evaluation rests on a 25-dataset collection with full provenance and a protocol fixed before results were read: matched thresholds, exact itemset-agreement gates, a 600 s timeout rule that records skips instead of hiding them, and 30-trial statistics on an 8-dataset subset. On 25 datasets, skarm's three en-

Table 6: Correctness verdicts: trial-0 itemset agreement per cell.

Cell	Pairs	Verdict	mlxtend	Note
accidents@0.5	1	all match	no	
accidents@0.6	4	all match	yes	
adult_census@0.05	4	all match	yes	
adult_census@0.1	4	all match	yes	
balance_scale@0.2	4	all match	yes	
balance_scale@0.3	4	all match	yes	
bms_pos@0.02	4	all match	yes	
bms_webview_1@0.005	4	all match	yes	
bms_webview_1@0.01	4	all match	yes	
bms_webview_2@0.005	4	all match	yes	
bms_webview_2@0.01	4	all match	yes	
car_evaluation@0.2	4	all match	yes	
car_evaluation@0.3	4	all match	yes	
chess@0.85	4	all match	yes	
connect@0.95	4	all match	yes	
docword_enron@0.5	3	all match	no	
docword_enron@0.7	3	all match	no	
docword_nips@0.5	4	all match	yes	
groceries@0.01	4	all match	yes	
groceries@0.02	4	all match	yes	
kosarak@0.005	1	all match	no	
kosarak@0.01	3	all match	no	
kosarak_spmf@0.01	3	all match	no	
market_basket@0.02	4	all match	yes	
market_basket@0.03	4	all match	yes	
mushroom@0.4	4	all match	yes	
online_retail@0.01	4	all match	yes	
online_retail@0.02	4	all match	yes	
online_retail_ii@0.01	4	all match	yes	
online_retail_ii@0.02	4	all match	yes	
pumsb@0.9	4	all match	yes	
pumsb_star@0.85	3	all match	no	
retail@0.02	4	all match	yes	
retail@0.03	4	all match	yes	
synthetic_dense_5000@0.3	4	all match	yes	
synthetic_dense_5000@0.5	4	all match	yes	
synthetic_sparse_2000@0.02	4	all match	yes	
synthetic_sparse_2000@0.05	4	all match	yes	
t10i4d100k@0.01	4	all match	yes	
t10i4d100k@0.02	4	all match	yes	
tic_tac_toe@0.3	4	all match	yes	
tic_tac_toe@0.5	4	all match	yes	

gines agree exactly on itemsets, apriori fastest on none; fpgrowth fastest on 5 of 25; eclat fastest on 20 of 25 at primary thresholds, with 6 timeout trials and mlxtend absent on 6 cells for stated reasons; the remaining work is the SPMF and modern-miner baselines, and the pruning-policy ablation.

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Table 7: Skipped, timed-out, and errored cells with reasons.

Dataset	Thr	Engine	Outcome	Reason
accidents	0.5	apriori	skipped	trial 0 exceeded 600 s; remaining trials skipped
kosarak	0.01	mlxtend	skipped	dense boolean DataFrame would need 40.9 GB
kosarak	0.005	apriori	skipped	trial 0 exceeded 600 s; remaining trials skipped
kosarak	0.005	mlxtend	skipped	dense boolean DataFrame would need 40.9 GB
kosarak_spmf	0.01	mlxtend	skipped	dense boolean DataFrame would need 5.3 GB
docword_enron	0.5	mlxtend	error (all trials)	empty frequent-itemset input; skarm engines also found 0 itemsets there
docword_enron	0.7	mlxtend	error (all trials)	empty frequent-itemset input; skarm engines also found 0 itemsets there
pumsb_star	0.85	mlxtend	error (all trials)	empty frequent-itemset input; skarm engines also found 0 itemsets there

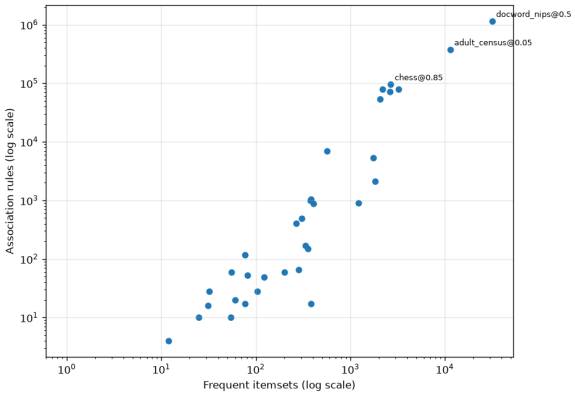


Figure 8: Frequent itemsets versus association rules per benchmark cell (log-log scale). Rule counts explode combinatorially ($2^k - 2$ candidates per itemset) on dense, long-transaction datasets, dominating total fit times. Six cells with zero rules are omitted.

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[14] scikit-learn developer documentation: estimator checks (`check_estimator`). `https://scikit-learn.org/stable/developers/develop.html` (accessed October 2026).

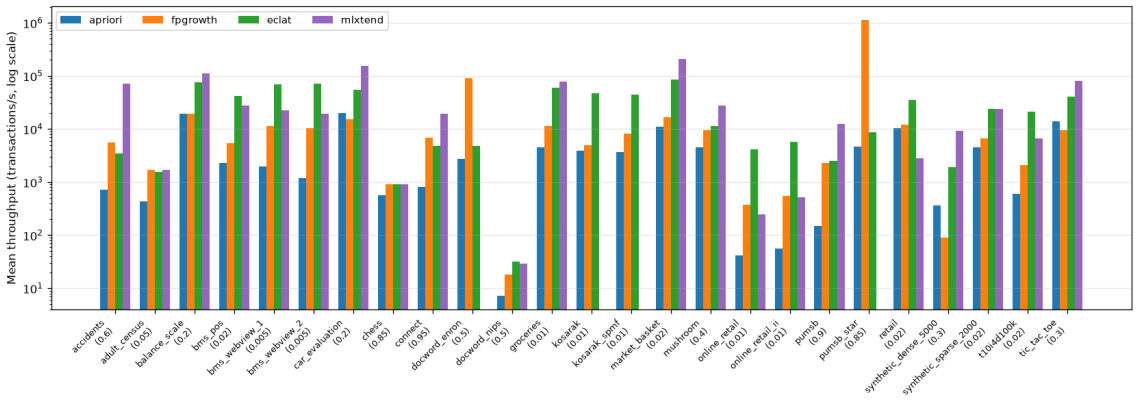


Figure 9: Mean throughput (transactions per second) per dataset at primary threshold (log scale). Reflects raw transactional processing rates; values favor short, highly structured transactions and reflect scale rather than intrinsic algorithm quality.

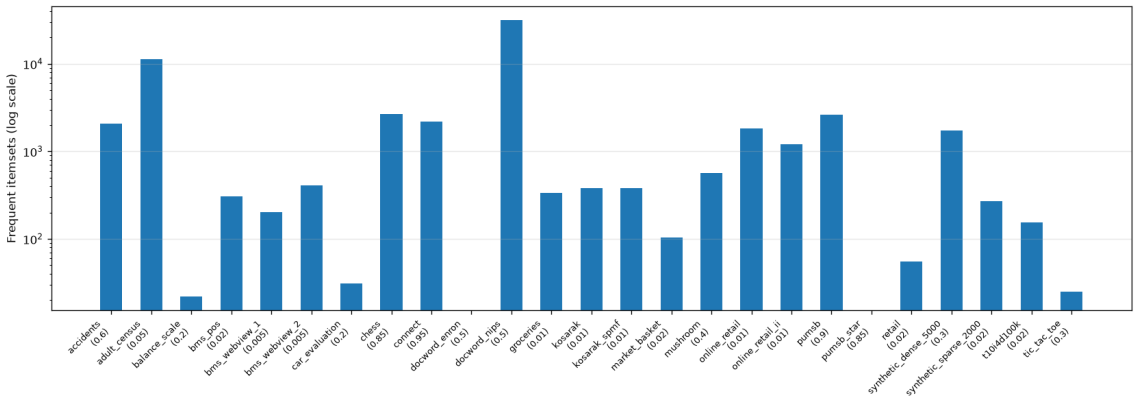


Figure 10: Frequent itemset scale across all 25 evaluation datasets at primary threshold (log scale). Output volumes span from a single frequent itemset (`tic_tac_toe@0.50`) to 31,582 itemsets (`docword_nips@0.50`), demonstrating the diversity of the benchmark suite.

A Reproducibility

Repository: `ABHISHEK1139/capstone` (main branch). Code fixes for the acceptance defects are commits `ff03b451`, `dd22782f`, `904b19b3`. Datasets live in `data/`; the full per-file manifest (source URL, access date 2026-10-29, licence, owner, exclusions, transformations) is `DATASET_INVENTORY.md`. The benchmark script is `benchmarks/run_pc_campaign.py` with thresholds in `benchmarks/thresholds.json`; the campaign is resume-safe via per-dataset checkpoint files, and trial records from different machines must never be merged into one results file. Results commit `79334222` (repository `benchmarks/campaign_results.json` plus `benchmarks/cells/` with 25 per-dataset checkpoint files). Machine and library versions are as in Section 5. The campaign report file (`benchmarks/CAMPAIGN_REPORT.md`) states COMPLETE. Note: `skarm/fpgrowth.py` in the repository carries a tail-pointer optimisation (O(1) node-link appends) added with the final two datasets; it changes no results (all correctness pairs agree exactly) and the methods section describes the code as it now stands. `benchmarks/thresholds.json` holds the matched ladder. Confirmed 2026-10-30 from the repository listing: `data/` holds 26 files — all 25 dataset files plus `generate_synthetic.py`. All 25 datasets in Table 3 are present; no data file is missing.

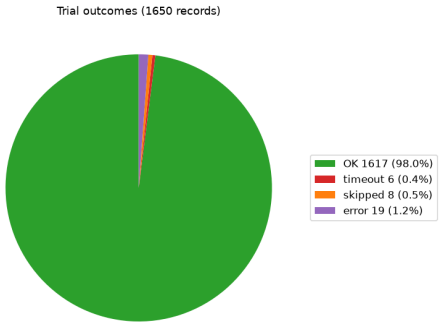


Figure 11: Statistical execution outcomes across all 1,650 trial records. The 98.0% completion rate (1,617 OK) certifies campaign robustness. Timeouts (0.36%), skips (0.48%), and `mlxtend` failures (1.15%) are audited with explicit rationale in Table 7.

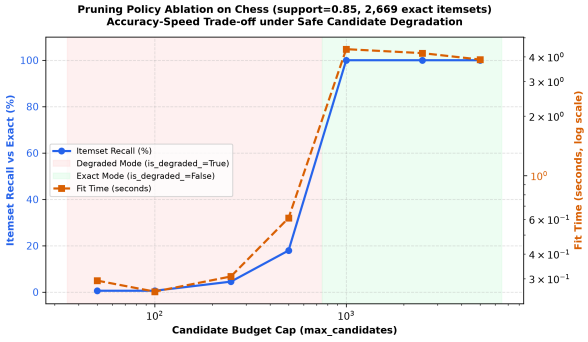


Figure 12: Pruning policy ablation on Chess ($N = 3{,}196$, minsup $= 0.85$, 2,669 exact frequent itemsets). Bounded mode halts combinatorial candidate explosion early, delivering up to 15.56× speedups at low caps while audibly signalling degradation via `is_degraded_`.

B Full cell-level timing table

One row per dataset × threshold × engine, over OK trials only; the Note column records failures for partial and empty cells, so no trial is hidden. This appendix exists so the main text can stay short; the condensed primary-threshold table is Table 5.

Table 8: Full per-cell timing and output counts. Mean, median (p50) and p95 are over OK trials only. Degraded mode never triggered on any cell, so no degraded column is shown.

Dataset	Thr	Engine	OK	Mean (s)	p50	p95	Peak MB	Items	Rules	Note
accidents	0.5	apriori	0	—	—	—	—	—	—	timeout 600 s
accidents	0.5	fpgrowth	5	156.5	157.1	181.3	955.1	8057	375774	—
accidents	0.5	eclat	5	288.2	273.6	333.4	1738.2	8057	375774	—
accidents	0.5	mlxtend	5	22.06	21.91	22.81	5494.2	8057	375774	—
accidents	0.6	apriori	5	473.5	475.2	478.2	1860.6	2074	54170	—
accidents	0.6	fpgrowth	5	60.92	61.31	62.38	90.38	2074	54170	—
accidents	0.6	eclat	5	97.93	97.65	100.3	1351.0	2074	54170	—
accidents	0.6	mlxtend	5	4.76	4.75	5.00	1470.8	2074	54170	—
adult_census	0.05	apriori	5	76.62	70.04	105.1	957.5	11390	376953	—
adult_census	0.05	fpgrowth	5	19.04	18.84	19.83	950.9	11390	376953	—
adult_census	0.05	eclat	5	20.95	20.34	24.01	887.9	11390	376953	—
adult_census	0.05	mlxtend	5	19.14	19.06	20.09	717.0	11390	376953	—
adult_census	0.1	apriori	5	23.36	20.41	31.18	160.6	3216	78167	—
adult_census	0.1	fpgrowth	5	5.53	5.55	5.58	150.6	3216	78167	—
adult_census	0.1	eclat	5	5.37	5.36	5.44	142.1	3216	78167	—
adult_census	0.1	mlxtend	5	3.48	3.48	3.55	208.8	3216	78167	—
balance_scale	0.2	apriori	5	0.033	0.032	0.039	0.064	22	0	—
balance_scale	0.2	fpgrowth	5	0.032	0.031	0.033	0.066	22	0	—
balance_scale	0.2	eclat	5	0.009	0.008	0.012	0.170	22	0	—
balance_scale	0.2	mlxtend	5	0.006	0.006	0.006	0.020	22	0	—
balance_scale	0.3	apriori	5	0.022	0.021	0.024	0.256	2	0	—
balance_scale	0.3	fpgrowth	5	0.022	0.022	0.024	0.018	2	0	—
balance_scale	0.3	eclat	5	0.008	0.008	0.009	0.024	2	0	—
balance_scale	0.3	mlxtend	5	0.017	0.006	0.062	0.210	2	0	—
bms_pos	0.02	apriori	30	227.2	232.6	237.2	22.79	308	489	—
bms_pos	0.02	fpgrowth	30	94.08	94.10	96.58	107.5	308	489	—
bms_pos	0.02	eclat	30	12.53	12.70	13.69	39.46	308	489	—
bms_pos	0.02	mlxtend	5	27.43	32.49	32.89	164.5	308	489	—
bms_webview_1	0.005	apriori	5	31.15	29.17	44.22	9.97	201	59	—
bms_webview_1	0.005	fpgrowth	5	5.31	5.24	6.58	2.92	201	59	—
bms_webview_1	0.005	eclat	5	0.860	0.875	0.985	8.52	201	59	—
bms_webview_1	0.005	mlxtend	5	2.68	2.51	3.62	1895.9	201	59	—
bms_webview_1	0.01	apriori	5	6.85	6.85	6.94	14.66	77	17	—
bms_webview_1	0.01	fpgrowth	5	2.30	2.29	2.36	2.47	77	17	—
bms_webview_1	0.01	eclat	5	0.377	0.379	0.388	8.23	77	17	—
bms_webview_1	0.01	mlxtend	5	0.397	0.409	0.411	401.3	77	17	—
bms_webview_2	0.005	apriori	5	64.30	63.04	70.42	14.63	408	877	—
bms_webview_2	0.005	fpgrowth	5	7.66	6.59	10.76	2.38	408	877	—
bms_webview_2	0.005	eclat	5	1.09	1.07	1.16	9.18	408	877	—
bms_webview_2	0.005	mlxtend	5	3.97	4.04	4.27	3423.4	408	877	—
bms_webview_2	0.01	apriori	5	12.51	11.03	16.67	16.28	81	52	—
bms_webview_2	0.01	fpgrowth	5	4.79	4.14	7.35	1.67	81	52	—
bms_webview_2	0.01	eclat	5	0.947	0.849	1.34	2.65	81	52	—
bms_webview_2	0.01	mlxtend	5	1.74	1.72	1.94	588.4	81	52	—
car_evaluation	0.2	apriori	5	0.087	0.087	0.088	0.104	31	16	—
car_evaluation	0.2	fpgrowth	5	0.112	0.110	0.119	0.260	31	16	—
car_evaluation	0.2	eclat	5	0.031	0.029	0.037	0.078	31	16	—
car_evaluation	0.2	mlxtend	5	0.011	0.011	0.012	0.016	31	16	—
car_evaluation	0.3	apriori	5	0.091	0.069	0.178	10.04	12	4	—
car_evaluation	0.3	fpgrowth	5	0.073	0.071	0.079	0.016	12	4	—
car_evaluation	0.3	eclat	5	0.026	0.026	0.027	0.060	12	4	—
car_evaluation	0.3	mlxtend	5	0.009	0.008	0.010	0.062	12	4	—
chess	0.85	apriori	30	5.76	5.75	6.96	167.4	2669	95482	—
chess	0.85	fpgrowth	30	3.49	3.45	3.85	105.7	2669	95482	—
chess	0.85	eclat	30	3.48	3.47	3.58	93.04	2669	95482	—
chess	0.85	mlxtend	5	3.50	3.52	3.57	19.94	2669	95482	—
connect	0.95	apriori	30	87.87	88.72	110.1	304.9	2201	78376	—
connect	0.95	fpgrowth	30	10.05	9.73	13.77	40.96	2201	78376	—
connect	0.95	eclat	30	14.22	14.30	15.30	191.0	2201	78376	—
connect	0.95	mlxtend	5	3.44	3.47	3.56	284.2	2201	78376	—
docword_enron	0.5	apriori	5	14.75	14.36	16.45	166.6	0	0	—
docword_enron	0.5	fpgrowth	5	0.446	0.445	0.455	7.64	0	0	—
docword_enron	0.5	eclat	5	8.37	8.04	9.49	17.74	0	0	—
docword_enron	0.5	mlxtend	0	—	—	—	—	—	—	mlxtend: empty input
docword_enron	0.7	apriori	5	14.25	14.24	14.36	191.1	0	0	—
docword_enron	0.7	fpgrowth	5	0.441	0.444	0.448	1.05	0	0	—
docword_enron	0.7	eclat	5	7.96	7.96	7.99	17.72	0	0	—
docword_enron	0.7	mlxtend	0	—	—	—	—	—	—	mlxtend: empty input

Dataset	Thr	Engine	OK	Mean (s)	p50	p95	Peak MB	Items	Rules	Note
docword_nips	0.5	apriori	30	209.5	196.2	269.3	2672.8	31582	1148468	—
docword_nips	0.5	fpgrowth	30	81.75	81.34	85.88	2560.6	31582	1148468	—
docword_nips	0.5	eclat	30	50.23	44.48	90.98	2345.5	31582	1148468	—
docword_nips	0.5	mlxtend	5	53.91	49.35	75.60	1079.8	31582	1148468	—
groceries	0.01	apriori	30	2.18	2.16	2.34	1.63	333	170	—
groceries	0.01	fpgrowth	30	0.871	0.871	0.938	1.15	333	170	—
groceries	0.01	eclat	30	0.166	0.160	0.203	0.588	333	170	—
groceries	0.01	mlxtend	5	0.128	0.126	0.142	107.0	333	170	—
groceries	0.02	apriori	30	1.07	1.07	1.13	0.841	122	49	—
groceries	0.02	fpgrowth	30	0.686	0.682	0.762	0.759	122	49	—
groceries	0.02	eclat	30	0.126	0.127	0.138	0.246	122	49	—
groceries	0.02	mlxtend	5	0.055	0.052	0.066	33.30	122	49	—
kosarak	0.005	apriori	0	—	—	—	—	—	—	timeout 600 s
kosarak	0.005	fpgrowth	5	473.9	477.5	481.2	69.29	1618	5890	—
kosarak	0.005	eclat	5	28.30	28.23	28.73	349.2	1618	5890	—
kosarak	0.01	apriori	5	250.8	249.9	253.1	690.8	383	1037	—
kosarak	0.01	fpgrowth	5	198.9	199.3	203.4	76.03	383	1037	—
kosarak	0.01	eclat	5	21.29	21.43	22.39	277.1	383	1037	—
kosarak_spmf	0.01	apriori	5	50.47	50.58	51.34	95.81	377	999	—
kosarak_spmf	0.01	fpgrowth	5	23.02	23.25	23.60	4.61	377	999	—
kosarak_spmf	0.01	eclat	5	4.24	4.25	4.32	25.35	377	999	—
market_basket	0.02	apriori	5	0.688	0.694	0.710	0.020	103	28	—
market_basket	0.02	fpgrowth	5	0.449	0.434	0.524	0.420	103	28	—
market_basket	0.02	eclat	5	0.089	0.088	0.102	0.020	103	28	—
market_basket	0.02	mlxtend	5	0.036	0.036	0.039	23.70	103	28	—
market_basket	0.03	apriori	5	0.449	0.425	0.556	0.026	54	10	—
market_basket	0.03	fpgrowth	5	0.370	0.368	0.401	0.426	54	10	—
market_basket	0.03	eclat	5	0.076	0.075	0.085	0.436	54	10	—
market_basket	0.03	mlxtend	5	0.023	0.020	0.029	8.73	54	10	—
mushroom	0.4	apriori	30	1.78	1.78	2.00	4.04	565	7020	—
mushroom	0.4	fpgrowth	30	0.849	0.848	0.900	1.97	565	7020	—
mushroom	0.4	eclat	30	0.718	0.714	0.777	2.52	565	7020	—
mushroom	0.4	mlxtend	5	0.298	0.292	0.338	3.91	565	7020	—
online_retail	0.01	apriori	5	496.5	495.0	539.7	48.06	1840	2108	—
online_retail	0.01	fpgrowth	5	55.79	55.89	56.52	19.52	1840	2108	—
online_retail	0.01	eclat	5	4.97	4.99	5.02	5.39	1840	2108	—
online_retail	0.01	mlxtend	5	85.08	82.62	108.5	9650.3	1840	2108	—
online_retail	0.02	apriori	5	70.87	71.01	73.18	32.43	353	151	—
online_retail	0.02	fpgrowth	5	17.99	18.07	18.17	4.78	353	151	—
online_retail	0.02	eclat	5	1.92	1.91	1.97	4.74	353	151	—
online_retail	0.02	mlxtend	5	1.99	1.98	2.12	2328.7	353	151	—
online_retail_ii	0.01	apriori	5	406.1	374.1	528.6	26.67	1216	911	—
online_retail_ii	0.01	fpgrowth	5	40.37	40.72	41.07	42.66	1216	911	—
online_retail_ii	0.01	eclat	5	3.90	3.93	3.96	6.40	1216	911	—
online_retail_ii	0.01	mlxtend	5	42.83	41.69	50.23	9785.3	1216	911	—
online_retail_ii	0.02	apriori	5	64.11	64.14	64.94	26.09	282	66	—
online_retail_ii	0.02	fpgrowth	5	15.78	15.63	16.49	3.16	282	66	—
online_retail_ii	0.02	eclat	5	1.81	1.81	1.84	6.15	282	66	—
online_retail_ii	0.02	mlxtend	5	2.30	2.15	2.67	1950.6	282	66	—
pumsb	0.9	apriori	5	332.8	323.1	374.8	87.70	2607	71474	—
pumsb	0.9	fpgrowth	5	21.34	21.23	21.74	25.96	2607	71474	—
pumsb	0.9	eclat	5	19.88	20.12	20.16	222.4	2607	71474	—
pumsb	0.9	mlxtend	5	3.90	3.97	4.08	397.6	2607	71474	—
pumsb_star	0.85	apriori	5	10.59	10.59	10.74	85.64	0	0	—
pumsb_star	0.85	fpgrowth	5	0.044	0.046	0.054	0.290	0	0	—
pumsb_star	0.85	eclat	5	5.69	5.70	5.81	17.49	0	0	—
pumsb_star	0.85	mlxtend	0	—	—	—	—	—	—	mlxtend: empty input
retail	0.02	apriori	30	8.44	8.43	8.65	30.09	55	60	—
retail	0.02	fpgrowth	30	7.28	7.28	7.46	0.559	55	60	—
retail	0.02	eclat	30	2.58	2.36	3.52	7.43	55	60	—
retail	0.02	mlxtend	5	31.90	32.20	36.84	2022.8	55	60	—
retail	0.03	apriori	30	7.29	7.28	7.41	31.77	32	28	—
retail	0.03	fpgrowth	30	7.21	7.23	7.58	1.06	32	28	—
retail	0.03	eclat	30	2.39	2.37	2.55	9.24	32	28	—
retail	0.03	mlxtend	5	15.00	14.95	15.34	2667.7	32	28	—
synthetic_dense_5000	0.3	apriori	5	13.67	13.62	14.23	4.93	1737	5262	—
synthetic_dense_5000	0.3	fpgrowth	5	55.95	57.17	58.87	16.32	1737	5262	—
synthetic_dense_5000	0.3	eclat	5	2.58	2.53	2.73	16.05	1737	5262	—
synthetic_dense_5000	0.3	mlxtend	5	0.551	0.529	0.603	380.1	1737	5262	—
synthetic_dense_5000	0.5	apriori	5	1.42	1.29	2.07	13.68	60	20	—
synthetic_dense_5000	0.5	fpgrowth	5	9.90	10.43	11.64	10.63	60	20	—
synthetic_dense_5000	0.5	eclat	5	0.547	0.545	0.590	9.27	60	20	—
synthetic_dense_5000	0.5	mlxtend	5	0.025	0.026	0.027	7.38	60	20	—
synthetic_sparse_2000	0.02	apriori	5	0.441	0.445	0.473	0.826	269	407	—
synthetic_sparse_2000	0.02	fpgrowth	5	0.325	0.277	0.544	12.73	269	407	—
synthetic_sparse_2000	0.02	eclat	5	0.085	0.089	0.097	0.264	269	407	—
synthetic_sparse_2000	0.02	mlxtend	5	0.086	0.090	0.100	3.46	269	407	—
synthetic_sparse_2000	0.05	apriori	5	0.190	0.195	0.202	0.974	77	118	—
synthetic_sparse_2000	0.05	fpgrowth	5	0.179	0.181	0.208	0.176	77	118	—
synthetic_sparse_2000	0.05	eclat	5	0.056	0.053	0.068	0.776	77	118	—

Dataset	Thr	Engine	OK	Mean (s)	p50	p95	Peak MB	Items	Rules	Note
synthetic_sparse_2000	0.05	mlxtend	5	0.034	0.035	0.039	0.836	77	118	—
t10i4d100k	0.01	apriori	1	592.9	592.9	592.9	0.000	385	17	4 failed: timeout 600 s
t10i4d100k	0.01	fpgrowth	5	72.63	71.45	77.40	292.7	385	17	—
t10i4d100k	0.01	eclat	5	6.49	6.39	7.53	49.25	385	17	—
t10i4d100k	0.01	mlxtend	1	101.9	101.9	101.9	0.000	385	17	4 failed: MemoryError: Unable to allocate 13.1 GiB
t10i4d100k	0.02	apriori	5	164.5	169.7	170.0	0.824	155	0	—
t10i4d100k	0.02	fpgrowth	5	48.41	48.07	49.81	124.5	155	0	—
t10i4d100k	0.02	eclat	5	4.78	4.80	5.01	23.61	155	0	—
t10i4d100k	0.02	mlxtend	5	34.40	40.95	42.95	17.31	155	0	—
tic_tac_toe	0.3	apriori	30	0.069	0.071	0.085	0.038	25	10	—
tic_tac_toe	0.3	fpgrowth	30	0.102	0.100	0.138	0.253	25	10	—
tic_tac_toe	0.3	eclat	30	0.024	0.023	0.031	0.059	25	10	—
tic_tac_toe	0.3	mlxtend	5	0.012	0.013	0.015	0.020	25	10	—
tic_tac_toe	0.5	apriori	30	0.046	0.046	0.058	0.025	1	0	—
tic_tac_toe	0.5	fpgrowth	30	0.045	0.044	0.064	0.021	1	0	—
tic_tac_toe	0.5	eclat	30	0.022	0.020	0.030	0.026	1	0	—
tic_tac_toe	0.5	mlxtend	5	0.005	0.005	0.006	0.040	1	0	—